Tendon-Driven Soft Arm: Static Positioning and Payload Compensation

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***Abstract*— A flexible inspection arm can bend to aim a camera or sensor, but the same tendon command may produce a different tip position after a payload is attached. This project will develop a planar, two-section tendon-driven arm simulator for static positioning under gravity. A discrete elastic beam will represent the backbone, while four tension-only tendons follow offset guides and anchor at the section junction or tip. The model will include stretching, bending, tendon elasticity, self-weight, and a tip payload. Implicit Euler steps with Newton iteration will relax the arm to equilibrium. A bounded inverse search will compare replayed unloaded commands with commands recomputed for the loaded arm. Analytical benchmarks, derivative checks, force balance, and mesh refinement will verify the implementation. The intended outcome is a reproducible comparison of positioning accuracy within a verified feasible workspace.**

I. Application and Objectives

Flexible inspection tools must orient a camera or sensor without bulky rigid joints. Continuum robots offer distributed bending and structural compliance, as discussed in soft-robotics and continuum-robot surveys [1], [2]. This project isolates one practical issue: attaching a tool changes gravitational loading, so commands calibrated without that tool may no longer reach the intended position.

The task is static tip positioning in a vertical plane. The project will determine (1) how tendon shortening and payload change equilibrium shape and reachable positions, and (2) whether load-aware commands reduce error relative to replaying unloaded commands. Comparisons will use identical targets and actuator limits. The contribution is an implemented and verified mechanics-based positioning study, rather than a new rod theory. Obstacle interaction and a complete inspection system are outside scope.

II. Related Work

Webster and Jones review constant-curvature kinematics for single- and multisection continuum robots [3]. This provides a useful geometric description, but the proposed model will determine curvature through force balance rather than prescribe it. Trivedi et al. include distributed weight and payload in geometrically exact soft-manipulator models [4]. Rucker and Webster couple tendon routing to continuum-robot mechanics under external loads [5], motivating explicit guide and anchor forces.

Bergou et al. present a discrete geometric treatment of elastic rods [6]. This project will use the planar beam counterpart taught in class. Renda et al. develop a different discrete Cosserat model for multisection soft arms [7], and Gravagne et al. study large-deflection planar continuum-robot dynamics and control [8]. These studies provide context; their full three-dimensional formulations and dynamic controllers will not be reproduced. The proposed scope retains tendon-induced section coupling and external loading while restricting the task to planar statics.

III. Physical System

A. Geometry and Degrees of Freedom

The initially straight arm points upward and has two equal sections, $L_1 = L_2 = L/2$. Centerline nodes $\boldsymbol{x}_i = (x_i, y_i)$ form the position vector $\boldsymbol{q}$. With N active edges and one extra edge inside the clamp, the system has $N + 2$ nodes. Prescribing the first two nodes at $(0, -h)$ and $(0, 0)$, where $h = L/N$, fixes root position and direction, leaving $2N$ free positional degrees of freedom. Free length L excludes the clamped edge. The section junction is part of a continuous backbone, not a hinge.

Each section has an opposing tendon pair. The proximal pair anchors at the junction; the distal pair passes through the proximal section and anchors at the tip (Fig. 1). Distal actuation therefore also loads the proximal section. Tendons slide through massless, frictionless guides at offset magnitude d. Guide positions and orientations follow the backbone, adding no independent degrees of freedom. Cross-section normals use normalized averages of adjacent edge tangents, with a one-sided tangent at the tip and a prescribed normal at the fixed root guide. The pairs represent independent tendon channels even where their planar projections coincide. Dense guiding is idealized: refining route samples approximates the same continuous offset path rather than changing the hardware.

B. Energies and External Loading

For current edge length ℓ_i and rest length ℓ_i^0,

$$E_{\text{arm}} = \sum_i \frac{EA}{2\ell_i^0}(\ell_i - \ell_i^0)^2 + \sum_i \frac{EI}{2\bar{\ell}_i}(\kappa_i - \kappa_i^0)^2. \qquad (1)$$

Here E, A, I are modulus, area, and second moment of area; $\bar{\ell}_i$ is the mean adjacent rest length. The discrete integrated curvature is $\kappa_i = 2\tan(\phi_i/2)$ for signed turning angle ϕ_i; initially $\kappa_i^0 = 0$. The backbone is geometrically nonlinear and materially linear elastic, with no shear or twist.

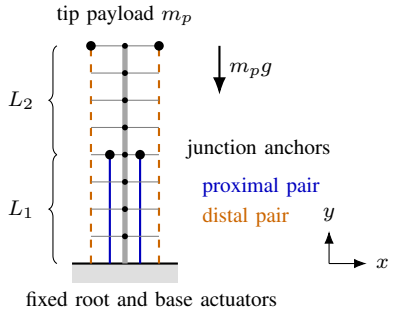


Fig. 1. Initial arm and routing. Tendon channels are separated for visibility; each has offset magnitude d. All tendons are base-driven. Gravity acts downward.

Tendon $a = (j, \sigma)$ is identified by its anchor section $j \in \{1, 2\}$ and side $\sigma \in \{-1, +1\}$. Its guide positions and routed length are

$$\boldsymbol{p}_{a,k} = \boldsymbol{x}_{i_k} + \sigma d\, \boldsymbol{n}_{i_k}, \qquad \ell_a(\boldsymbol{q}) = \sum_k \|\boldsymbol{p}_{a,k+1} - \boldsymbol{p}_{a,k}\|. \quad (2)$$

Each route starts at its fixed base guide and ends at its anchor. The commanded free length is $\ell_a^u = \ell_a^{\text{init}} - \sigma u_j$, where ℓ_a^{init} is the initial routed length and u_j is signed differential shortening. For $\langle z \rangle_+ = \max(z, 0)$,

$$E_a = \tfrac{1}{2} K_a \langle \ell_a - \ell_a^u \rangle_+^2, \qquad T_a = K_a \langle \ell_a - \ell_a^u \rangle_+, \quad (3)$$

where K_a is tendon axial stiffness in N/m. A tendon can slacken but never carries compression. Frictionless sliding gives uniform tension along each tendon. Differentiating its complete path includes forces and moments at guides and anchors. On a taut branch,

$$\boldsymbol{F}_a = -T_a \nabla \ell_a, \qquad \nabla^2 E_a = K_a \nabla \ell_a \nabla \ell_a^T + T_a \nabla^2 \ell_a. \quad (4)$$

Slack tendons contribute zero, and the active set is reevaluated during iteration.

With upward y, total potential is

$$\Pi(\boldsymbol{q}, \boldsymbol{u}) = E_{\text{arm}} + \sum_a E_a + \sum_i m_i g y_i + m_p g y_{\text{tip}}. \quad (5)$$

The payload also augments tip mass during relaxation. Tendon friction, contact, and out-of-plane motion are excluded.

C. Starting Parameters

Table I lists numerical design choices, not measured hardware properties. Pilot runs will check the operating range before the comparison protocol is frozen. The circular backbone has $A = \pi r^2$ and $I = \pi r^4/4$.

IV. Simulation and Positioning Algorithms

A. Forward Equilibrium

The implementation will adapt the course's planar beam derivatives and free-coordinate assembly. Automatic differentiation will evaluate routing derivatives on each taut/slack branch, checked independently against finite differences on small problems.

TABLE I
Proposed Starting Values

Quantity	Value or range
Free length; section lengths	0.20 m; 0.10 m each
Radius; effective modulus	6 mm; 10 MPa
Density; gravity	1000 kg/m^3; 9.81 m/s^2
Tendon offset; stiffness	8 mm; 1000 N/m
Commands; payload	$u_j \in [-4, 4]$ mm; $m_p = 0, 2, 5$ g
Active edges	40; refinement at 20, 80
Initial relaxation step	$\Delta t = 0.01$ s

Algorithm 1 Forward Simulation to Static Equilibrium

Require: Commands $\boldsymbol{u}$, payload m_p, initial state
for each command and load increment **do**
  repeat
    Save state; initialize Newton guess
    repeat
      Evaluate guide paths and taut/slack tendons
      Assemble $\nabla\Pi, \nabla^2\Pi, \boldsymbol{R}, \boldsymbol{J}$
      Solve $\boldsymbol{J}_{ff} \Delta \boldsymbol{q}_f = -\boldsymbol{R}_f$
      Backtrack and update free coordinates
    until step converges or fails
    On failure, reduce Δt and retry
    Update velocity from the accepted step
  until static force and motion criteria both hold
end for
Check strain, tensions, geometry, and stability
return Equilibrium, tip position, and status

Damped relaxation solves $\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \nabla\Pi = \boldsymbol{0}$ with numerical damping $\boldsymbol{C} = \gamma \boldsymbol{M}$. Implicit Euler gives

$$\begin{aligned} \boldsymbol{R} = {} & \frac{\boldsymbol{M}}{\Delta t^2}(\boldsymbol{q}^{n+1} - \boldsymbol{q}^n - \Delta t\, \boldsymbol{v}^n) \\ & + \frac{\boldsymbol{C}}{\Delta t}(\boldsymbol{q}^{n+1} - \boldsymbol{q}^n) + \nabla\Pi(\boldsymbol{q}^{n+1}, \boldsymbol{u}), \end{aligned} \quad (6)$$

with Newton matrix $\boldsymbol{J} = \boldsymbol{M}/\Delta t^2 + \boldsymbol{C}/\Delta t + \nabla^2\Pi$. Backtracking will limit invalid geometry and reduce the residual. Failed steps are retried at smaller Δt; commands are ramped from accepted states. Iteration and relaxation-step limits will return an explicit failure status rather than allow indefinite runs.

Acceptance requires free static force residual below $10^{-5} F_{\text{ref}}$ and maximum nodal change below $10^{-5} L$ for 20 consecutive steps, with decaying velocity. Define $F_{\text{ref}} = \max(EI/L^2, m_{\text{total}} g, 10^{-3}\,\text{N})$. Here m_{total} includes backbone and payload mass. The free potential Hessian will be checked for stability; switching states will receive additional perturbation checks. Numerical damping is solely a route to static equilibrium. Physical settling time and damping are not evaluation outputs.

B. Inverse Positioning and Comparison

Let $\boldsymbol{p}(\boldsymbol{u}, m_p)$ be the converged tip position. Bounded nonlinear least squares will minimize

$$\frac{\|\boldsymbol{p}(\boldsymbol{u}, m_p) - \boldsymbol{p}^*\|^2}{L^2} + \lambda \frac{\|\boldsymbol{u}\|^2}{u_{\max}^2}, \qquad |u_j| \le u_{\max}, \quad (7)$$

initially using $\lambda = 10^{-6}$. Finite differences of converged forward solves give tip sensitivities. Several starting commands from a coarse scan will reduce local-minimum sensitivity.

A 9×9 command scan and local refinement will identify stable operating regions. Ten to fifteen non-collinear targets will be chosen in the common verified workspace of unloaded and loaded arms. Nearly singular command-to-tip mappings, especially near the straight pose, will be flagged. Provisional limits of 2 N tendon tension and 2% backbone axial strain will exclude invalid states, as will self-intersection or nearly reversed adjacent edges. Limits and target rules will remain fixed during the comparison.

For each target, unloaded command $\boldsymbol{u}_0$ will be replayed on the loaded arm, then a load-aware command $\boldsymbol{u}_{m_p}$ will be computed. Errors are

$$\begin{aligned} e_{\text{replay}} &= \|\boldsymbol{p}(\boldsymbol{u}_0, m_p) - \boldsymbol{p}^*\|, \\ e_{\text{comp}} &= \|\boldsymbol{p}(\boldsymbol{u}_{m_p}, m_p) - \boldsymbol{p}^*\|. \end{aligned} \tag{8}$$

Outputs will include mean and maximum error, successful-target fraction, tensions, and solve cost. An error below 1% of arm length is a proposed goal, not a claimed result. Commands will also be evaluated on a finer mesh. Forward failures will be recorded rather than interpreted as valid positions. The payload is known; the task is model-based command compensation, not online load estimation.

V. Validation and Expected Results

Verification will precede positioning experiments.

1) *Beam benchmarks:* Disable tendons and gravity. For a small transverse tip force F, compare displacement with $FL^3/(3EI)$. In a separate small end-moment test, compare rotation and displacement with $ML/(EI)$ and $ML^2/(2EI)$.
2) *Tendon and derivative checks:* Verify zero compression force, $T = K_a \Delta\ell$ on a fixed straight route, and virtual work $\delta E_a = T_a \delta\ell_a$ at fixed command. Check energy gradients and Hessian-vector products by centered differences away from taut/slack transitions.
3) *Balance and refinement:* Check root reactions, gravity, and base-tendon forces, including moment balance. Refine active edges from 20 to 40 to 80 at fixed physical dimensions. Proposed goals are less than $0.01L$ tip-position change and 5% taut-tendon-tension change between the last two meshes. Repeat representative cases with halved time steps and different numerical damping.

Expected outputs are shape animations, workspace maps, tendon-tension curves, and paired error plots. The hypothesis is that payload changes the command-to-position mapping and that load-aware commands reduce error for feasible targets. Small improvements and failed cases will be reported. These tests verify implementation; quantitative prediction for real hardware would require material calibration and experiments.

VI. Work Plan and Deliverables

Stage 1 will implement a single section, tendon routing, and analytical and derivative tests. Stage 2 will add the distal pair, verify coupling, and complete mesh and workspace studies. Stage 3 will perform inverse positioning and payload comparisons, followed by the report and presentation. Single-section operation is an intermediate milestone; the final goal remains the two-section arm.

Deliverables will include Python forward and positioning solvers, reproducible validation cases, saved comparison data and figures, and a README. Adapted course routines will retain attribution. The report will state assumptions and unsuccessful cases. Hardware fabrication, friction, contact, three-dimensional motion, and dynamic tracking are excluded from the required scope.

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